Association of adenovirus infection with human obesity.

We previously reported that chickens infected with the avian adenovirus SMAM-1 developed a unique syndrome characterized by excessive intra-abdominal fat deposition accompanied by paradoxically low serum cholesterol and triglyceride levels. There have been no previous reports of avian adenoviruses infecting humans. We screened the serum of 52 humans with obesity in Bombay, India, for antibodies against SMAM-1 virus using the agar gel precipitation test (AGPT) method. Bodyweights and serum cholesterol and triglyceride levels were compared in SMAM-1-positive (P-AGPT) and SMAM-1-negative (N-AGPT) groups. Ten subjects were positive for antibodies to SMAM-1, and 42 subjects did not have antibodies. The P-AGPT group had a significantly higher bodyweight ($p < 0.02$) and body mass index ($p < 0.001$) (95.1 +/- 2.1 kg and 35.3 +/- 1.5 kg/m2, respectively) compared with the N-AGPT group (80.1 +/- 0.6 kg and 30.7 +/- 0.6 kg/m2, respectively). Also, the P-AGPT group had significantly lower serum cholesterol ($p < 0.02$) and triglyceride ($p < 0.001$) values (4.65 mmol/L and 1.45 mmol/L, respectively) compared with the N-AGPT group (5.51 mmol/L and 2.44 mmol/L, respectively). Two subjects positive for SMAM-1 antibodies had antibodies against each others' serum, suggesting the presence of antigens in one or both. When these two serum samples were inoculated into chicken embryos, macroscopic lesions compatible with SMAM-1 infection developed. The inoculation of serum from N-AGPT subjects did not produce such lesions. The presence of increased obesity, antibodies to SMAM-1, reduced levels of blood lipids, and viremia that produces a typical infection in chicken embryos suggests that SMAM-1, or a serologically similar human virus, may be involved in the cause of obesity in some humans.